TING-YU (JOHAN) LU

Mechanical & Marine Engineer | Technical Project Manager | 5 Patents

Relocating to Stockholm, August 2026 | **Full work authorization — no sponsorship required**

PROFILE

Entrepreneurial R&D engineer with 14 years of experience in offshore marine engineering, amphibious vehicle development, cleantech water treatment, and hardware prototyping. Led a 20-person cross-functional R&D team, secured USD 1.2M in government R&D funding, and hold 5 granted patents (water treatment systems, amphibious suspension, UAV control). Concurrently founded a STEM education institution guiding 200+ students to top universities, including coaching national rocketry competition teams praised by expert panels as *"exceeding typical high school maturity."*

KEY EXPERIENCE

Deputy Engineer & Project Manager 2013 – 2016

Ship and Ocean Industries R&D Center (SOIC) — National Research Institute, Taipei

- Led 20-person R&D team through full development cycle of an amphibious vehicle; secured TWD 36M (~USD 1.2M) government funding and 2 patents.
- Managed offshore wind turbine seafastening platform development, coordinating 20+ suppliers to DNV/GL classification standards (MSC/NASTRAN, ABAQUS analysis).
- Supported submarine R&D: torpedo launch system design and underwater explosion simulation modeling.
- Completed 30+ yacht inspections and certifications; advised 3 startups on yacht service ventures.

Engineering Director & Partner 2016 – Present

KINDAX Mechanical (Water Treatment) & NEMOR Co. Ltd (Product Design), Taipei

- Developed sustainable aquaculture aeration and filtration systems, resulting in 3 granted patents (2024–2025) for microbubble diffuser and water treatment technologies.
- Managed 20+ hardware prototyping projects (USD 500K+ combined), delivering 100+ product units from concept to batch production for commercial clients.

Founder & STEM Program Director 2016 – Present

Mathematics & Science Engineering Education Institution, Taitung

- Built and operated a STEM centre serving 120+ students/year; 200+ alumni admitted to top universities (16 to medical/dental programs).
- Coached 2 national rocketry teams (TASA Taiwan Cup): students designed dual-redundancy avionics, custom PCBs, and FMEA-driven safety protocols.
- Developed AI-assisted university admission analytics tool (Python/SQLite), serving 100+ candidates annually.

PATENTS (5 GRANTED)

Microbubble Aeration Pipe & Device (I909845, 2025) | **Filtration Device & Filter Tube** (M676719, 2025) | **Aeration Device** (M658218, 2024)

Foldable Suspension System for Amphibious Vehicle (I592319, 2017) | **UAV Autopilot Control for Ultra-Light Aircraft** (M423667, 2012)

EDUCATION

M.E. Aeronautic & Astronautic Engineering — National Cheng Kung University (NCKU) 2010–2012

B.E. Mechanical Engineering — National Taiwan University (NTU) 2006–2010

TECHNICAL SKILLS & LANGUAGES

Engineering: SolidWorks, CATIA, MSC/NASTRAN, ABAQUS, CFD, 3D Printing/Rapid Prototyping

Software: Python, C++, SQLite, Data Analysis, AI Application Development

Certifications: Licensed Yacht Captain (Taiwan Class 2), Power Boat License

Languages: Mandarin (Native), English (Professional), Swedish (Beginner — actively studying)

PUBLICATIONS

Lu, T.-Y. *et al.*, *"System Reengineering in Mechanism Design for Unmanned Ultra-Light,"* J. Aeronautics, Astronautics and Aviation, 46(2), 2014.

Chang, W.-C., **Lu, T.-Y.** *et al.*, *"Strength Analysis of Offshore Wind Turbine Seafastening Fixtures,"* 28th CSNAME Conference, NTU, 2016.

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